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18 June 1964
SUPERSEDING
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(See 6.3)

MILITARY SPECIFICATION

COMPASS, GYRO, MINIATURE

1. SCOPE

1.1 This specification covers the requirements for the design, manufacture, examination, and test of a miniature gyro compass to furnish true heading indications.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-S-901 - Shock Tests, HI (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-I-983 - Interior Communication Equipment, Naval Shipboard; Basic Design Requirements for.
MIL-C-5015 - Connectors, Electric, "AN" Type.
MIL-Q-9858 - Quality Program Requirements.
MIL-M-15071 - Manuals, Equipment and Systems.
MIL-B-15072 - Batteries, Storage, Lead-Acid, Portable Except for Aircraft and Automotive Vehicles.
MIL-A-15303 - Audible Signals: Alarms, Bells, Buzzers, Horns and Sirens - Shipboard.
MIL-M-17413 - Motor, Direct-Current, Integral - HP (Shipboard Use).
MIL-E-17555 - Electronic and Electrical Equipment and Associated Repair Parts - Preparation for Delivery of.
MIL-S-20708 - Synchros, 60 and 400 Cycle, General Specification.

STANDARD

MILITARY

MIL-STD-710 - Synchros, 60 and 400 Cycle.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 When specified (see 6.1), a preproduction sample of the miniature gyro compass equipment shall be furnished for the tests specified in 4.3 (see 6.2).

3.2 Description.- The miniature gyro compass equipment shall consist of the following:

(a) Compass unit.
(b) Control unit (may be integral with the compass unit).
(c) Speed unit.
(d) Alarm unit (for single or multiple compass installation (see 6.1)).

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(e) Visual alarm (see 6.1).
(f) Isolation transformer.
(g) Power supply unit (see 6.1).

3.3 General features.- In addition to the requirements specified herein, the equipment shall be in accordance with MIL-I-983. (Whenever a requirement of MIL-I-983 conflicts with a requirement of this specification, the requirements of this specification shall govern.)

3.4 Material.- All parts, except those where strength or hardness is essential, shall be constructed of nonferrous, nonmagnetic materials.

3.5 Design.- The miniature gyro compass equipment shall be designed to operate from a 3-phase, 115-volt, 400-cycle supply with tolerances specified in MIL-I-983. Total 400-cycle power requirements to operate the compass equipment excluding the power supply unit shall not exceed 400 volt amperes. Cable entrance into enclosures may be by stuffing tubes or AN type connectors in accordance with MIL-C-5015. All enclosures shall be dripproof. The compass equipment shall be designed so as not to be damaged by failure of power to the equipment. All components shall be able to pass the vital equipment vibration requirements of MIL-I-983 as specified herein. All components shall be able to pass the type A high shock requirements of MIL-S-901 for lightweight equipment as specified herein. Internal resilient mounts may be used in the compass unit or in an integral compass and control unit only to meet the shock requirement. External resilient mounts shall not be used.

3.5.1 Compass unit.- The compass unit shall, by means of a gravity controlled gyroscope responsive to earth rates, continuously seek and indicate the true meridian. The true heading of the carrier (a vehicle or ship) shall be readable from an integral dial of the compass unit. This unit shall be provided with a protective enclosure designed for three point deck or shelf mounting. Mounting holes shall provide for ± 5 degrees adjustment. A fore and aft line so marked shall be engraved on the base of the enclosure to facilitate alignment to the carrier's centerline. The fore and aft alignment marks shall be referred to the innermost heading reference alignment mark within ± 6 minutes of arc. The compass unit shall perform as a gravity controlled gyroscope at any latitude between 75 degrees south and 75 degrees north. At latitudes over 75 degrees to the poles, the compass unit shall perform as a dependable direction indicating gyro with a drift rate not in excess of 0.6 degree per hour.

3.5.1.1 The compass unit shall be so designed that it will indicate true heading with an error not to exceed ± 1.5 degrees when the unit is subjected to a magnetic field of 10 gausses.

3.5.1.2 Maximum rate of turn.- The follow-up system of the compass unit shall be capable of turning in azimuth at any rate up to 100 degrees a second. The follow-up shall be fitted with alarm contacts which will indicate failure of the follow-up system by means of an indicator lamp on the control unit. The compass equipment shall not be susceptible to damage upon failure of the follow-up system. The follow-up amplifier shall meet the test specified in 4.6.9.

3.5.1.3 Settling.- Means shall be provided for electrically leveling and slewing the gyroscopic element of the compass unit so that it can be leveled and set within 10 degrees of the true heading. A tilt indicator shall be provided either on the compass unit or on the control unit. The compass unit shall be capable of meeting the settling test specified in 4.6.1.

3.5.1.4 Noise.- The gyro compass equipment shall withstand the tests for airborne audible, structure-borne and radio frequency noise specified in 4.6.13.

3.5.1.5 Bearings.- The gyro rotor or rotors shall be designed for a minimum life expectancy of 15,000 hours.

3.5.1.6 Rotor vibration.- The gyro rotor or rotors shall be designed to operate from zero speed to 15 percent overspeed without any critical periods of vibration in the operating range or any objectionable critical periods throughout the entire range.

3.5.1.7 Roll and pitch clearance.- The compass unit shall be operable under conditions of a roll of 45 degrees or a pitch of 45 degrees on each side of the vertical.

3.5.1.8 The effect of the carrier's speed and course on the true heading indication shall be automatically compensated for the gyroscopic element to indicate the true meridian instead of the dynamic meridian. Effects

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of change of latitude on the true heading indication shall be compensated for by the manual introduction of latitude signals in the control unit.

3.5.1.9 Pitch acceleration.- The compass unit shall be so designed that it will not be adversely affected by pitch accelerations up to 1,000 degrees per second and pitch velocities up to 50 degrees per second. In addition, it shall not be adversely affected by a fore and aft horizontal acceleration of 1 g of 2 seconds duration or by a momentary fore and aft horizontal acceleration of 3 g's.

3.5.1.10 The compass unit shall transmit heading at 1 and 36 speed. The synchro transmitters shall be 18TR4 standard Navy synchros in accordance with MIL-S-20708 and MIL-STD-710.

3.5.1.11 Provision shall be made for preventing the unauthorized changing of the compass unit adjustments.

3.5.1.12 The volume of the compass unit with or without an integral control unit shall not exceed a nominal 2 cubic feet.

3.5.2 Control unit.- The control unit shall contain the amplifiers, computer circuitry, power supplies, meters, alarm indicator lights and switches to control and monitor compass operation. The enclosure for the control unit, unless made integral with the compass unit, shall be designed for three point bulkhead mounting. The volume of the control unit shall not exceed a nominal 1-1/4 cubic feet.

3.5.2.1 Accessibility.- The power supplies and amplifiers shall be capable of being removed for examination and of being tested with plug and jack connections. Test cables 10 feet long fitted with proper connectors to allow the removable subassembly to be electrically connected to the control unit shall be included unless the front cover of the control unit is arranged to serve as a service platform. In the latter case, the plug half of the connector may be fitted with a shorter electrical cable which will allow servicing the subassemblies on the front cover.

3.5.2.2 Operating knobs.- Operating knobs on the control panel shall be so constructed or protected that accidental contact will not disturb them or change their setting. Introduction of latitude compensation signals shall be by a knob and scale on this unit.

3.5.2.3 Instruction plates.- Instruction plates may be placed on the control unit where necessary.

3.5.2.4 Electron tubes and solid state semi-conductor devices.- For power rectification, solid state semi-conductor devices shall be used instead of electron tubes.

3.5.3 Speed unit.- The speed unit shall, to facilitate making the speed compensations specified in 3.5.1.8, accept a 60-cycle, 0 to 40 knot synchro input where 9 degrees of synchro rotation equal one knot. The speed unit shall generate a signal proportional to speed as required by the design of the compass equipment. This unit shall be enclosed in an enclosure suited for three point bulkhead mounting.

3.5.3.1 Provision shall be made to manually insert the speed input, using a dial calibrated 0 to 40 knots.

3.5.3.2 The size of the speed unit shall not exceed a nominal one-eighth cubic foot.

3.5.4 Alarm unit.- The alarm unit shall consist of necessary relays and components to actuate one or more audible alarms like the B5DSF4 bell of MIL-A-15303 and one or more visual alarms in the event of failure in a synchro signal amplifier, which is used with, but is not a part of, the compass equipment, or when certain critical circuits or functions of the compass equipment become inoperative, operate under par, or when there is a power failure. For a multiple compass installation the alarm unit shall contain the necessary components to, in addition to the above, actuate the audible or visual alarms upon closure of an external alarm circuit in any one of the compass systems installed.

3.5.4.1 The size of the alarm unit shall not exceed a nominal 1/6 cubic foot. This unit shall be constructed for three point bulkhead mounting.

3.5.5 Visual alarm.- The visual alarm shall contain one or more lights. A steady light shall indicate normal functioning of the compass(s), power supply or synchro signal amplifier; a flashing light shall indicate an alarm condition in a compass(s), in a synchro signal amplifier or failure of the power supply.

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3.5.5.1 The visual alarm shall be no larger than a nominal 1/6 cubic foot. The alarm shall be constructed for three point bulkhead mounting.

3.5.6 Isolation transformer.- The isolation transformer shall be provided for interposing between the carrier's 115 volt, 3-phase, 400-cycle supply and the compass equipment.

3.5.7 Power supply unit.- A static type power supply unit and appropriate controls shall be furnished to provide automatically 400-cycle power to the compass and loads on the compass transmitters to keep the compass and those loads operating normally during periods of ship's supply failure or intolerably low voltage or low frequency in the ship's supply. The unit and its controls shall be designed so that for ship's voltage and frequency values below the lower limits in MIL-I-983 the unit will operate from a 120 volt direct current source (type BB255/U, BB252/U, or BB256/U batteries of MIL-B-15072 or a rectifier). The unit shall be automatically switched to charge the batteries upon restoration of normal ship's power. The unit shall be suited for continuous operation when operating from ship's supply or from a direct current supply. (When used with a rectifier the power supply unit shall not furnish power to or through the rectifier to other loads on the rectifier.) The unit shall not be damaged if connected for an indefinite period of time to partially or fully discharged batteries and the compass equipment shall not be damaged when dependent on the power supply unit during this time. Protection against damage from discharged batteries shall not be dependent on fuses or circuit breakers. The unit in addition to supplying the power required to operate the compass equipment shall be capable of supplying an additional 250-watt, 0.8 power factor (p.f.), 1-phase, 400-cycle external equipment load. The total harmonic content of the output of the unit shall not be more than 5 percent of the fundamental and the maximum of any one harmonic shall not be more than 2 percent of the fundamental. The unit shall be no less than 60 percent efficient when operating from the ship's supply and no less than 70 percent efficient when operating from a direct current supply. Voltage and frequency regulation when the unit is operating from direct current in the range 105-120 volts shall be ± 2 volts and ± 2 cycles per second (c.p.s.) for no load to full load. The unit shall provide a battery charging voltage of 129 volts ± 2 volts when operating from any ship's supply voltage and frequency within the limits of MIL-I-983. Facility shall be built into the unit wherewith to set the battery charging current rate within the following limits for the specific type of battery being used of those listed above:

	Charging rates in amperes	
Battery type	Start	Finish
BB255/U	1.5	1
BB252/U	8	4
BB256/U	14	7

The power supply unit shall be designed so that for line-to-line faults or 3-phase line-to-line faults for a duration of at least 5 seconds no fuses or circuit breakers shall blow or open, no damage shall be done to the unit and upon clearing the fault the unit shall continue to operate as designed. The size of the power supply unit shall not exceed 5 cubic feet. The unit shall be designed for bulkhead mounting and shall be designed for a minimum of 5000 hours mean time between failures.

3.5.8 Interchangeability.- Equipment shall be so designated that any compass unit of the same mark and modification will operate as specified in this specification with any control and amplifier unit or any speed transmitting unit of the corresponding mark and modification without requiring any modification other than minor adjustment of electrical circuits such as are practicable aboard ship.

3.5.9 Temperature and humidity.- The compass equipment shall operate within the accuracy limits specified in 3.5.9.1 at any temperature from 40°F. to 149°F. when tested in accordance with 4.6.3. If necessary, an integral heater or blower may be provided to accomplish this requirement. The equipment shall be protected for failure of the heater or blower.

3.5.9.1 At temperatures from 40°F. to 110°F. the turntable error of 3.5.10 shall be allowed.

3.5.9.2 Extended storage.- The equipment shall be suitable for the extended storage requirements of MIL-I-983.

3.5.10 The limits of allowable errors for the gyro compass equipment shall be as shown in table I for the latitude of the contractor's plant or that of the commercial laboratory and are referred to the 1-speed synchro transmitter. The limits include synchro transmitter errors, settling point error and allowable alignment errors.

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Table I - Limits of allowable errors.

Name of test	Allowable errors in degrees (±)
Settling point (difference)	0.2
Settling point	0.75
Supply line voltage and frequency variation	1.0
Turntable	0.75
Small angle Scorsby	1.0
Large angle Scorsby	2.0
Effect of magnetic fields	1.5
Correction for speed and latitude	0.15

3.5.11 Service record book.- The supplier shall record the following data in the Service Record Book, NAVSHIPS 708, which is Government-furnished:

Total test time in hours compass has sustained _____ .
Date compass was accepted for delivery _____ .
Damping _____ percent.
Compass period _____ minutes.
Average settling point during settling tests _____ .
Serial number of gyro assembly _____ .
Settling time _____ .
Test procedure for shipboard personnel to recheck the above data.

3.5.12 Drawings.- Drawings shall be as specified in MIL-I-983. For any component of the compass equipment weighing 100 pounds or more the installation drawing shall include the estimated weight and vertical location of the center of gravity. The final installation drawing shall include the actual weight and actual vertical location of the center of gravity.

3.5.13 Manuals.- Manuals shall be type I of MIL-M-15071 and the quantity required shall be as specified (see 6.1).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification; where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality program requirements.- The supplier shall provide and maintain a quality program acceptable to the Government. The program shall be in accordance with MIL-Q-9858. The supplier is in no way relieved of the final responsibility to furnish gyro compass equipment meeting the requirements of this specification.

4.2 General procedures.- The procedures, definitions and conditions applicable for inspection to be performed on the equipment shall be as specified hereinafter. Except where specifically indicated, all tests shall be performed at room temperature (nominally 75°F.) with normal supply voltage and frequency, with the speed and latitude settings which prevail and with the compass operating in its normal mode of operation for latitudes 0-75 degrees. All readings shall be recorded in degrees, tenths of a degree and to the nearest hundredth of a degree. Unless otherwise stated, readings of compass indications for all tests shall be taken at 15 minute intervals. Furthermore, for each test the compass shall be aligned to a reference heading, which reference heading shall be accurate to within ± 30 seconds. The compass shall be aligned in azimuth to the reference heading within ± 3 minutes. The compass error shall be measured either by reading out the 36-speed compass transmitter outputs on a 36-speed dial of a gyro repeater or recorder and comparing the reading with the Scorsby reading, or by using a synchro control transformer geared at 36-speed to the Scorsby test stand, furnishing signals from the 36-speed compass transmitters to the control transformers and reading the control

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transformer rotor voltage on a calibrated meter or recorder. Unless specific test headings are designated, the selected headings shall have the concurrence of the Government representative. The compass binnacle or output transmitters of the compass shall not be shifted to center any settling point error.

4.2.1 Equipment mounting for inspection requirements.-

4.2.1.1 Equipment.- The compass equipment shall be tested on a MK5 or MK6 or equivalent Scorsby test stand.

4.2.1.2 The compass unit shall be mounted on the Scorsby test plate, or on the turntable, so that the fore and aft line of the compass enclosure base lies in the vertical plane of the fore and aft line of the turntable, or Scorsby test plate. The fore and aft line of the Scorsby test plate shall be so oriented that the gimbal axes of the gyro compass will always remain parallel to the corresponding axes of the Scorsby test stand.

4.2.1.3 After the compass unit has been mounted on the Scorsby test plate or turntable, no change shall be made affecting the position of the compass unit with respect to the Scorsby test plate or turntable, or in the lubber line of the compass unit, to correct for change in the settling point of the compass unit.

4.2.2 Definitions.-

4.2.2.1 Test stand attitude.-

4.2.2.2 Test stand heading.- The test stand heading is defined as the angle between a north and south vertical plane and the vertical plane through the fore and aft line of the Scorsby test plate, measured in the horizontal plane in a clockwise direction from north.

4.2.2.3 Test stand roll.- The test stand roll is defined as the angle between the plane of the testplate and the horizontal plane, measured in the plane perpendicular to the fore and aft line of the test plate; positive when starboard side of test plate is up.

4.2.2.4 Test stand pitch.- The test stand pitch is defined as the angle between the plane of the test plate and the horizontal plane, measured in a vertical plane through the fore and aft line of the test plate; positive when the forward end of the test plate is down.

4.2.2.5 Compass unit error.- The compass unit error is defined as the angular difference between the indications of the compass 36-speed transmitter and the test stand attitude. The compass error is positive when the compass transmitter indication is greater than the test stand indication.

4.2.2.6 Settling point.- The settling point of a compass unit on test is defined as the average of three consecutive readings of the compass unit taken at 15-minute intervals during which the compass error at no time exceeds the settling point error specified in 3.5.10 and during which the difference between any two of these readings does not exceed the settling point difference in 3.5.10. For settling point determination the correction devices shall be adjusted for zero speed at the local latitude setting. The compass is considered to have settled at the start of the first reading from which the settling point is obtained, that is, at the time at which the error becomes (and from which the error subsequently remains) less than the allowed settling point error.

4.3 Preproduction inspection.- Preproduction inspection shall be conducted, when specified (see 3.1 and 6.1). Preproduction equipment shall be subjected to the examination and tests listed in table II.

Table II - Preproduction inspection.

Examination and tests	Requirement reference	Test reference
General examination	MIL-I-983	4.5
Enclosure	3.5	MIL-I-983
Settling point	3.5.1.3 and 3.5.10	4.6.1
Damping	3.5.11	4.6.4
Turntable	3.5.10	4.6.6
Maximum rate of turn	3.5.1.2	4.6.17

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Table II - Preproduction inspection (Cont'd).

Examination and tests	Requirement reference	Test reference
Supply line voltage and frequency variation	3.5.10	4.6.2
Angular freedom	3.5.1.7	4.6.18
Small angle Scorsby test	3.5.10	4.6.7
Interrupted power test	3.5.7	4.6.8
Power transfer test	3.5.7	4.6.8.1
Follow-up amplifier failure test	3.5.1.2	4.6.9
Alarm unit test	3.5.4	4.6.19
Large angle Scorsby test	3.5.10	4.6.10
Correction for speed and latitude test	3.5.1.8 and 3.5.10	4.6.11
Directional gyro test	3.5.1	4.6.5
Noise	3.5.1.4 and MIL-I-983	4.6.13
Effect of magnetic field	3.5.1.1	4.6.15
Vibration	3.5	4.6.16
Temperature and humidity	3.5.9	4.6.3
Rotor vibration test	3.5.1.6	4.6.12
Dielectric strength	MIL-I-983	4.6.14.1
Insulation resistance	MIL-I-983	4.6.14.2
Shock	3.5	4.6.16

4.4 Quality conformance inspection.- All equipment shall be subjected to the examination and tests listed in table III which shall be performed generally in the order listed.

Table III - Quality conformance inspection.

Examination and tests	Requirement reference	Test paragraph
General examination	MIL-I-983	4.5
Dielectric strength	MIL-I-983	4.6.14.1
Insulation resistance	MIL-I-983	4.6.14.2
Settling point	3.5.1.3 and 3.5.10	4.6.1
Damping	3.5.11	4.6.4
Turntable	3.5.10	4.6.6
Small angle Scorsby test	3.5.10	4.6.7
Interrupted power test	3.5.7	4.6.8
Power transfer test	3.5.7	4.6.8.1
Follow-up amplifier failure test	3.5.1.2	4.6.9
Alarm unit test	3.5.4	4.6.19
Large angle Scorsby test	3.5.10	4.6.10
Correction for speed and latitude test	3.5.1.8 and 3.5.10	4.6.11
Directional gyro test	3.5.1	4.6.5

4.5 General examination.- The general examination shall be conducted as specified in MIL-I-983. No operating test is required as part of the general examination.

4.6 Test procedures.-

4.6.1 Settling.- The gyro compass shall have the capability of settling with an error no greater than the settling point error of 3.5.10 on any heading according to the conditions of 4.2.2.6 and the following:

4.6.1.1 Settling readings shall be taken at 5-minute intervals near the peaks and at 1-minute intervals on the peaks whenever the compass unit is required to settle in preparation for any part of the test. During the settling period the compass unit shall be kept stationary (except for the dynamic settling test) and vertical on the Scorsby test stand, and the correction devices shall be adjusted for zero speed at the local latitude setting.

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4.6.1.2 At the beginning of any part of the test, if the last readings for the preceding part show that the compass unit has settled as specified in 4.2.2.6, the settling readings may be waived for the part of the test being started, unless settling readings are requested by the supplier. However, the settling readings shall not be continued beyond the time at which the compass unit has first settled as specified in 4.2.2.6, unless such settling readings are required for conducting the test.

4.6.1.3 Rapid settling.- When preset to within 10 degrees of the true heading, the compass unit shall come to and indicate within 1.5 degrees of the true heading in less than 1 hour when started at a nominal temperature of 20°F. with the compass unit stationary.

4.6.1.4 Dynamic settling.- The compass shall have the capability to settle in less than 4 hours with an error no greater than the settling point error of 3.5.10 when started up and run under large angle Scorsby test motions at a nominal room temperature of 75°F. and when set within 10 degrees of the true heading.

4.6.1.5 Time to settle.- Not more than 4 hours shall be allowed at any time for the compass unit to settle for any set of test conditions specified herein.

4.6.2 Supply line voltage and frequency variation.- The gyro compass shall, after being settled in accordance with 4.2.2.6 with voltage at 115 volts and frequency at 400 c.p.s., sustain a sudden change to the power supply voltage and frequency limits specified herein, and shall meet therewith and thereafter the accuracy requirements of 3.5.10 on any heading while undergoing a turntable test and on any heading while undergoing the small and large angle Scorsby tests.

(a) Frequency at 380 c.p.s., voltage at 109 volts, with the frequency not changing more than 2 c.p.s. per hour after reaching 380 c.p.s.
(b) Frequency at 420 c.p.s., voltage at 121 volts, with the frequency not changing more than 2.c.p.s. per hour after reaching 420 c.p.s.
(c) Frequency and voltage several times varying between the extremes of the power supply limits allowed by MIL-I-983 with the variation occurring in some instances suddenly and in other instances more slowly.

(A test on only one heading for the above test condition is adequate.)

Note: For the above tests, readings shall be taken at least every 15 minutes.

4.6.3 Temperature and humidity test.- For this test the compass shall be settled according to 4.2.2.6 on one heading. The following procedure shall apply with the compass operating in its normal mode. The compass shall at no time exceed the turntable error of 3.5.10 at temperatures from 40° to 110°F. and shall at no time exceed twice the turntable error of 3.5.10 for higher temperatures. The power supply unit shall be tested at the extreme temperatures of this test after its temperature has stabilized to demonstrate that it will meet the requirements of 3.5.7.

Hours	Ambient temperature and relative humidity during test	Angle of tilt
With the compass at an equilibrium temperature of 40°F., start and settle the compass. This shall require no more than 4 hours. Record the time required.		
0-21	40°F./95 percent relative humidity.	Normal
21-27	Raise the temperature in 6 hours to 149°F., maintain 95 percent relative humidity.	Normal
27-152	Temperature 149°F., relative humidity 95 percent at alternate 1 hour periods, that is, 1 hour at 95 percent relative humidity and the next hour with relative humidity uncontrolled. Measure air temperature external and internal to the enclosure and check parts for hot spots, at the end of 125 hours.	Normal
152-156	Lower the temperature and humidity to room conditions, nominally 75°F. and 50 to 60 percent relative humidity.	Normal

Throughout the above test the compass error shall be recorded either continuously or at regular 15-minute intervals. Hot spot temperatures shall not exceed the maximum dictated by affected insulation as specified in MIL-I-983 and shall be performed according to MIL-I-983. The maximum internal temperature rise shall be

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recorded and shall not exceed 90°F. from the ambient. Equipment installation and the air change rate capability of the test chamber shall be in accordance with MIL-I-983.

4.6.3.1 Extended storage.- For this test the procedure shall be in accordance with MIL-I-983. The equipment after stabilizing at the room temperature shall be capable of settling with an error no greater than the settling point error of 3.5.10.

4.6.4 Damping test.-

4.6.4.1 The compass unit shall be settled as specified in 4.2.2.6, on any heading designated by the supplier, with concurrence of cognizant Government representative. The sensitive element shall then be carefully precessed 5 degrees from the established meridian and leveled so that the level reading is in such direction as to increase the deflection from the established meridian. The compass unit shall then be allowed to settle as specified in 4.2.2.6. The time required to settle shall be not more than 4 hours.

4.6.4.2 The mean percentage of damping is:

$$\frac{(a) - (c)}{(a) + (b)} \times 100$$

where:

a = Amplitude first peak.
b = Amplitude second peak.
c = Amplitude third peak.

4.6.4.3 The mean percentage of damping shall not differ more than plus or minus 10 percent from the value given by the supplier in accordance with 3.5.11 and shall be between 60 and 80 percent.

4.6.5 Directional gyro test.-

4.6.5.1 The compass unit shall be settled as specified in 4.2.2.6 on any heading designated by the supplier with the concurrence of the cognizant Government representative. The compass shall then be operated in the directional gyro mode for a 2-hour period. The drift rate for this test shall not exceed 0.6 degree per hour.

4.6.6 Turntable test.-

4.6.6.1 The compass unit shall be mounted on a Scorsby test stand and settled as specified in 4.2.2.6. It shall then be given a 4-hour test on each of four headings of the Scorsby test stand as designated by the supplier with the concurrence of cognizant Government representative.

4.6.6.2 The compass unit error shall at no time during this test exceed the limit specified in 3.5.10 for turntable.

4.6.7 Small angle Scorsby test.-

4.6.7.1 The compass unit shall be mounted on a Scorsby test stand and shall be settled as specified in 4.2.2.6. It shall then be given an 8-hour test on each of the following six headings:

(a) Any two cardinal headings (0, 90, 180 and 270 degrees).
(b) Any two intercardinal headings (45, 135, 225, and 315 degrees).
(c) Any two semi-intercardinal headings (22.5, 67.5, 112.5, 157.5, 202.5, 247.5, 292.5, and 337.5 degrees).

4.6.7.2 On each heading the sequence of test shall be as follows:

Time	Condition
4 hours	Roll, pitch and yaw
2 hours	Roll only
2 hours	Pitch only

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4.6.7.3 The supplier with concurrence of the cognizant Government representative shall designate the sequence of the headings which are to be used during this test. The amplitude of roll on each side of the vertical shall be 15 degrees, with a period between 8 and 10 seconds. The amplitude of pitch on each side of the vertical shall be 5 degrees, with a period between 6 and 8 seconds. The amplitude of yaw on each side of the Scorsby test stand heading shall be 5 degrees with a period between 8 and 10 seconds.

4.6.7.4 The compass unit error shall not exceed the limit specified in 3.5.10 for this test.

4.6.8 Interrupted power test.- This test shall be conducted as specified in 4.6.8.1 except that ship's power shall be interrupted instead of reduced gradually. The allowable error shall be that specified in 3.5.10 for the large angle Scorsby test.

4.6.8.1 Power transfer test.- The compass shall be settled in accordance with 4.2.2.6 on any heading designated by the supplier with the concurrence of the cognizant Government representative. The compass shall then be operated under large angle Scorsby conditions on one heading. The normal power supply shall then be reduced to a voltage or frequency less than the lower ship's supply limits allowed by MIL-I-983. Transfer of operation to the power supply unit shall be effected automatically. The compass shall operate on the power supply unit for several minutes and then with ship's power restored to normal voltage and frequency shall transfer back to ship's power automatically. The compass error as shown by readings taken for 1 hour or more beginning 5 minutes after transfer back to ship's power shall not exceed the allowable error specified in 3.5.10 for the large angle Scorsby test.

4.6.9 Follow-up amplifier failure test.- The compass unit shall be settled in accordance with 4.2.2.6 on any heading selected by the supplier with concurrence of the cognizant Government representative. Then operation of the amplifier in the follow-up system shall be interrupted for a period of 5 minutes, simulating a failure. The Scorsby turntable shall then be operated under large angle conditions. The amplifier shall then be reconnected and the compass unit shall be capable of normal functioning.

4.6.10 Large angle Scorsby test.- This test shall be conducted as specified in 4.6.7 for the small angle Scorsby test, except that the amplitude of roll shall be 40 degrees on each side of the vertical, with a period between 8 and 10 seconds; the amplitude of pitch shall be 10 degrees on each side of the vertical, with a period between 6 and 8 seconds; and the amplitude of yaw shall be 10 degrees on each side of the Scorsby test stand heading with a period of 10 to 12 seconds. The compass unit error shall not exceed the limit specified in 3.5.10.

4.6.11 Correction for speed and latitude test.-

4.6.11.1 The compass unit shall be settled as specified in 4.2.2.6 on any heading designated by the supplier with concurrence of cognizant Government representative.

4.6.11.2 The speed and latitude correction shall be set for zero speed and local latitude, and the Scorsby turntable shall be rotated to a heading designated by the supplier with concurrence of cognizant Government representative. The correction shall then be set for a specified speed and the compass allowed to settle. The true heading indication of the compass unit shall be recorded for the zero speed input and the specific speed input.

4.6.11.3 The procedure specified in 4.6.11.1 shall be repeated for several well-distributed combinations of speed and heading.

4.6.11.4 The indicated corrections obtained shall be compared with the values computed by the following formula:

$$\tan C = \frac{S \cos C^1}{e \cos L - S \sin C^1}$$

where:

C = Virtual meridian correction.
S = Speed setting in knots.
C^1 = Course setting (heading).
e = 902.46 knots.
L = Local latitude in degrees.

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The difference between the indicated correction and the corresponding computed correction shall not in any case exceed the allowable error specified in 3.5.10 for this test.

4.6.11.5 For the "correction for speed and latitude test," the compass equipment shall be operated from the ship's underwater log or from a device which simulates the output of the log equipment.

4.6.12 Rotor vibration test.- The gyro rotor or rotors shall be operated from zero speed to normal operating speed, and shall be operated as specified in 4.6.12.1 and 4.6.12.2. The rotor or rotors shall have no critical periods of vibration in the operating range, nor any objectionable critical periods throughout the entire range at any time during this test. Temperature rise shall not exceed that specified in 4.6.12.1 and 4.6.12.2.

4.6.12.1 Each gyro rotor, if designed for vacuum or gas, shall be run in its case at an overspeed of at least 15 percent for a period of not less than 1 hour, during which period there shall be no objectionable vibration or noise of the gyro rotor, and the temperature rise of the bearings shall not exceed 67°C. above the ambient (room) temperature, as determined by the thermometer method specified in MIL-M-17413 or any other method satisfactory to the Government representative.

4.6.12.2 Where the design does not involve hermetic sealing and requires gyro rotor or rotors to operate a vacuum or gas, the rotor or rotors shall be capable of being operated in air at atmospheric pressure with a 40-percent reduction in speed for 4 hours without exceeding a temperature rise of 67°C. above the ambient (room) temperature, as determined by the thermometer method specified in MIL-M-17413 or any other method satisfactory to the Government representative.

4.6.13 Noise.- Each unit of the compass equipment shall be tested separately for compliance with the radio frequency noise requirements of 3.5.1.4 and for compliance with the airborne audible and structure-borne noise requirements of 3.5.1.4 for other than low-noise applications unless requirements for low-noise applications are specified (see 6.1).

4.6.14 Dielectric strength and insulation resistance tests.-

4.6.14.1 Dielectric strength.- Dielectric strength between electrical circuits and between electrical circuits and ground shall be tested in accordance with MIL-I-983. For preproduction tests, the qualification inspection conditions of MIL-I-983 shall apply; otherwise the acceptance inspection conditions shall apply. In the case of low voltage components, waivers shall be obtained in specific instances permitting dielectric tests to be made at less than 600 volts in consideration of reasons of compelling importance. Gyro assemblies shall not be given this test.

4.6.14.2 Insulation resistance.- Immediately after passing the dielectric test, each circuit of each instrument or unit shall be tested for insulation resistance in accordance with MIL-I-983. A potential of 500 volts, d.c. shall be applied between a terminal and the ground when the temperature of the instrument is approximately the normal operating temperature.

4.6.15 Effect of magnetic field.- The compass unit with its binnacle shall be given a test to determine the effect of degaussing. The effect of magnetic fields on the compass unit shall be determined by using Helmholtz coils capable of producing a magnetic field of 10 gausses uniform intensity and direction over a cube 3 feet on a side when the permeability of the cube is 1. No adjustment of the field strength shall be made after the insertion of the equipment between the coils. The compass shall be mounted midway between the coils with the gyro rotor approximately on the axis of the coils, and shall be settled with no field applied. The field shall be applied in any direction, the compass unit allowed to settle, and the maximum error and new settling position noted. In no case shall the settling with the field applied exceed 4 hours. The direction of the magnetic field shall then be reversed and the procedure repeated. These two tests shall be repeated for each of the two directions perpendicular to the original direction of the magnetic field and for a direction of the magnetic field which shall be inclined 45 degrees in the vertical plane parallel to the gyro spin axis. The maximum compass error shall not exceed the value specified in 3.5.10.

4.6.16 Shock and vibration.- One of each component of the equipment shall be subjected to the shock and vibration tests to determine conformance to 3.3 and 3.5. The compass unit shall be settled in accordance with 4.2.2.6 before the test. During the vibration test the compass unit shall be required at any time to not exceed an error of ± 0.75 degrees. During the shock test the compass unit shall be required at any time to not exceed an error of ± 4 degrees. After both the shock and vibration tests the compass unit shall be capable of settling

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with an error not in excess of the settling error allowed in 3.5.10. All other units of the compass equipment shall, to meet the shock and vibration tests, not suffer a failure which would interfere with proper compass operation during or after the test.

4.6.17 Maximum rate of turn.- The compass unit shall be settled as specified in 4.2.2.6 on any heading designated by the Government representative. The turntable shall be turned in azimuth in the following manner: The platform shall be smoothly accelerated from rest to a velocity of 100 degrees per second, rotated at a constant velocity of 100 degrees per second for approximately one revolution, and then smoothly decelerated to rest. During the constant velocity part of the motion of the overall lag shall be determined by energizing from the gyro compass, a control transformer driven by the turntable and measuring the output voltage. The actual lag shall be computed from the control transformer calibration, and shall not exceed 15 minutes of arc. The compass unit error as determined from the output of the calibrated control transformer at 5-minute intervals for 30 minutes, beginning 5 minutes after the rotation, shall not exceed 1 degree.

4.6.18 Angular freedom.- The test for angular freedom shall be performed as follows: Settle the compass as specified in 4.2.2.6 on one heading. Thereupon tilt the turntable 45 degrees to port in roll and maintain the turntable in this attitude approximately 30 minutes, then repeat for a 45-degree roll to starboard. Repeat this procedure three times. Thereafter do the same procedure for a 45-degree downward pitch and a 45-degree upward pitch. The roll and pitch attitudes should be obtained at slow rates, approximately 10 degrees per second. At no time during the test should the compass become unsettled.

4.6.19 Test of alarm system.- The alarm system shall be tested to determine conformance with 3.5.4.

4.7 Spare sensitive element.- When a spare gyroscopic element assembly is provided in the contract, it shall be tested in any compass unit of that same contract selected by the supplier with concurrence of the Government representative. The inspection shall consist of the following tests and shall be conducted under the same conditions as for the compass unit:

(a) Rotor vibration (see 4.6.12).
(b) Damping (see 4.6.4).
(c) Turntable (see 4.6.6).
(d) Small angle Scorsby (see 4.6.7).
(e) Insulation (see 4.6.14).

4.8 Failure.-

4.8.1 The occurrence of any of the following conditions shall require the subsequent running of the turntable test, small angle Scorsby test, large angle Scorsby test, and such other tests as the supplier with concurrence of cognizant Government representative may consider to be affected by the failure or the subsequent adjustments:

(a) Compass unit error in excess of the allowable limits during the turntable test, small angle Scorsby test, large angle Scorsby test, except as provided in 4.8.3.
(b) Failure to settle as specified in 4.2.2.6 or failure to remain settled at any time during the test.
(c) Failure, defective operation, breakdown on the insulation test, or need for adjustment of any part of the gyroscopic element or follow-up system, including amplifier, at any time during the test.
(d) Loss of rotor case vacuum.

4.8.2 Except as provided in 4.8.1, any failure of the compass unit to conform to the specified accuracy or other limit of operation or a failure of any auxiliary component of the equipment affecting the operation of the compass unit shall require the subsequent running of those tests which the supplier with concurrence of cognizant Government representative may consider to be affected by the failure of the subsequent adjustments.

4.8.3 Trial rerun.- In case the compass unit error exceeds the allowed limit during the turntable test, small angle Scorsby test, or large angle Scorsby test, and there is no apparent cause for this condition, the contractor may request a rerun of those tests on which the excessive error was observed, for the purpose of determining the cause of the error. If the reasons for such a run are satisfactory, the rerun shall be made under carefully observed conditions. If the results of the rerun are satisfactory, the test in question shall be considered complete and the excessive error during the original run shall not be considered a failure.

4.8.4 A failure of any auxiliary unit of the equipment (except parts of the gyroscopic element and follow-up system) which affects the operation of the compass unit, while being tested as specified for quality assurance, shall require the rerunning of the test on which the failure occurred.

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5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking.- The equipment and accessories, repair parts and technical publications shall be preserved and packaged level A or C; packed level A or B as specified (see 6.1) and marked in accordance with MIL-E-17555. Method II shall apply for level A preservation and packaging of equipment.

5.2 One Government furnished Service Record Book, NAVSHIPS 708, shall be packed with each gyro compass.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Quantity of items required.
(c) Whether preproduction sample is required (see 3.1 and 4.3).
(d) Whether alarm unit is for single or multiple compass installation (see 3.2).
(e) If visual bridge alarm indicator is required with each compass (see 3.2).
(f) Whether power supply unit is required with each compass (see 3.2).
(g) Inventory control point for repair parts.
(h) Quantity of repair parts.
(i) Type and quantity of drawings required (see 3.5.12).
(j) Quantity of type I manuals required (see 3.5.13).
(k) Grade C equipment for low noise applications, both for airborne audible and structureborne noise 3.5.1.4 and 4.6.13).
(l) Time contractor is to retain inspection records for Government examination.
(m) Level of preservation, packaging and packing required (see 5.1).

6.2 Preproduction.- Invitations for bids should provide that the Government reserves the right to waive the requirement for preproduction samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

6.3 CHANGES FROM PREVIOUS ISSUE. THE EXTENT OF CHANGES (DELETIONS, ADDITION, ETC.) PRECLUDE THE ANNOTATION OF THE INDIVIDUAL CHANGES FROM THE PREVIOUS ISSUE OF THIS DOCUMENT.

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT

$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426

REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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